

MID Server Best Practices

Whitepaper

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Application and Architecture

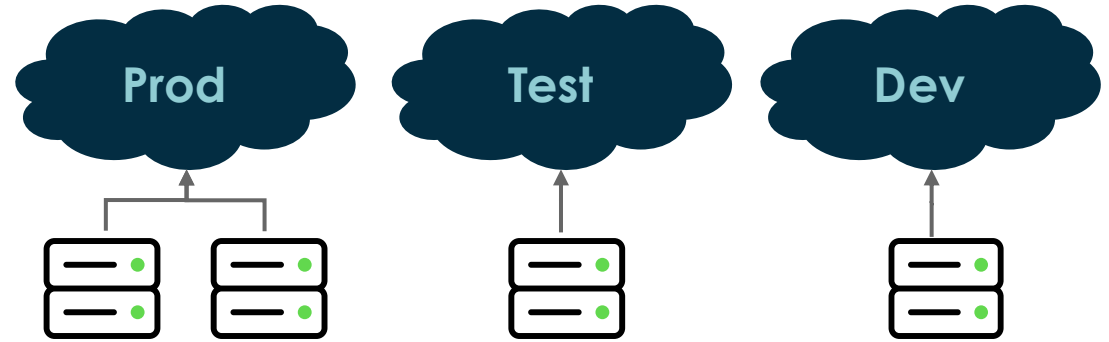


MID Servers

- Lightweight Java application that runs as a **Windows service** or UNIX daemon
- **Must** be hosted on Windows to Discover Windows & Unix.
- Set the supported application to the Application you are using, for example set the Application to "Discovery" if using the MID Server for Discovery
- MID Servers cannot be shared between instances
- Job is to execute jobs and return the results back to the instance for processing.
- It is recommended to cluster 'like' MID Servers (covered in later slides)
- Secure **outbound only** 128/256-bit SSL communication to your specific ServiceNow applications



More Information



- Deploy the MID Server application on a local drive on the host within a unique folder name with no space in it (i.e. c:\ServiceNow\MID_Server_Agent)
- Redundancy is needed in production environments
- May be relaxed in sub-production
- Consider using 2048 MB of JVM memory
- Set the tuning to 125 threads
- Start planning production deployment
 - Clustering, Performance, Access
 - Use results from scans during development

MID Server Requirements

- Windows Server 2012 or greater
 - .NET Framework 3.5, 4.0, 4.5, 4.6, or 4.7
 - PowerShell version 3.0 and supports versions up to PowerShell 5.1
- Linux Redhat 6+
- Ubuntu 14+
- CentOS 6+

- Standard Hard drive partition ~40GB
- 8GB Memory
- Quad Core processor 2+ GHz
- 64-bit systems
- Memory required can range from 4GB to 16GB

- Ensure a validated connection from the servers to be monitored and the MID server

- For Event Management – Check to see if you need the MID Server Extension enabled.

Consider any change requests that need to be submitted to have the MID Servers built



More Information

Number of MID Servers required

- MID Server performance and system requirement are driven by multiple factors including but not limited to:
 - The size of the customer infrastructure and specific function (Capability) MID Server is configured to perform.
 - The ITOM Application you are deploying.
 - The number of MID Servers installed on the host.
 - The number of threads per MID Server.
 - The number and nature of other applications running on MID Server host.
 - Architectural and design enhancements over ServiceNow releases.
 - Consider access, location and availability
 - For Discovery – Use the [MID Server Calculator](#)



**How many
MID servers
are
needed?**

**Start
with at
least 2**

Leading Practice:

Production environment, you should configure at least two MID Servers to support zero-touch configuration and ensure that a MID Server is always available if one fails

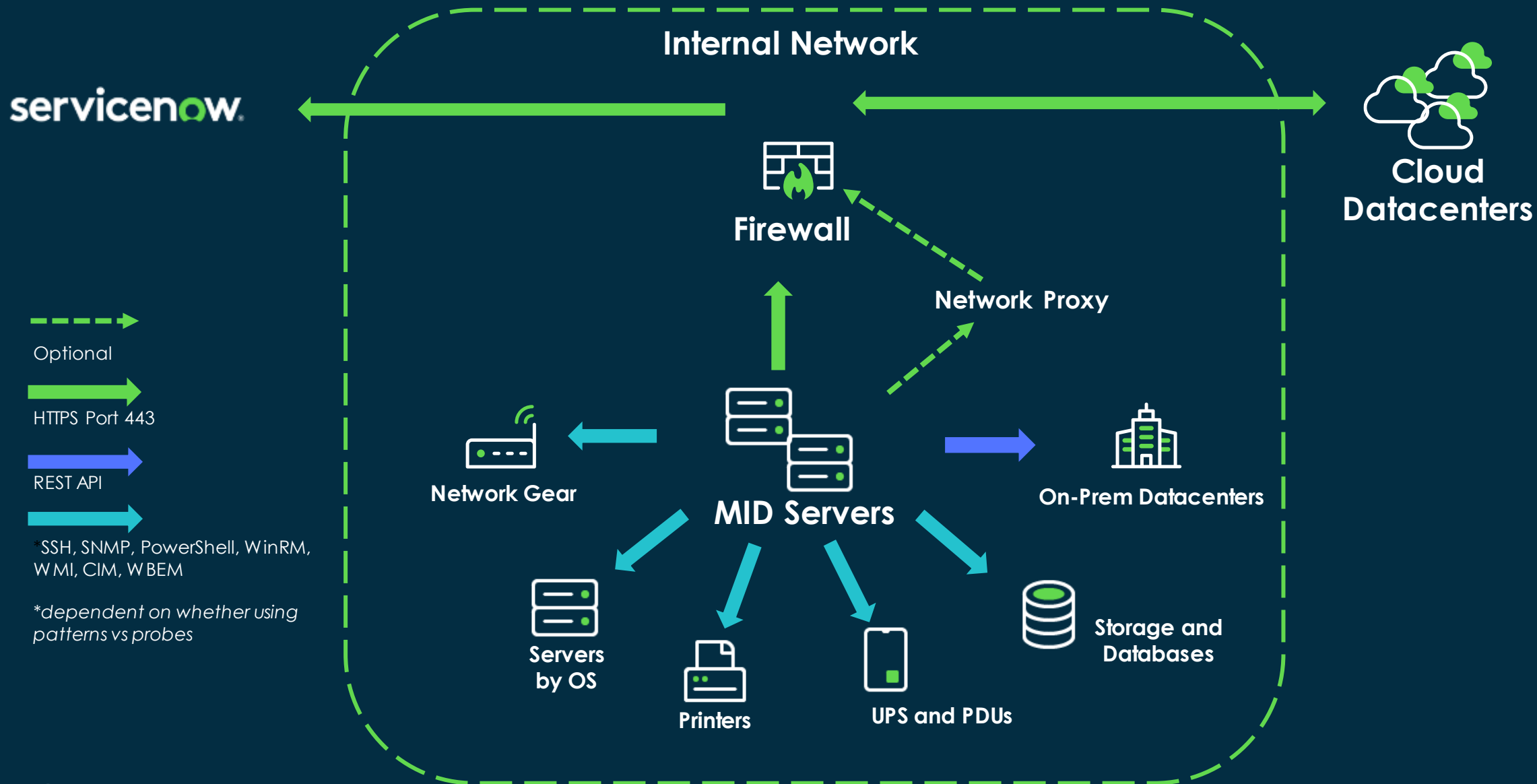
Clustered MID Servers

- Multiple MID servers should be setup in the network assigned to the same IP Address ranges.
- When MIDs are setup in Clustered groups, ServiceNow automatically balances the load on Event Management across the different MID Servers and they act as a failover for each other in the event one MID Server goes down.
- Adding an additional MID Server is a good leading practice when there are large numbers of devices (e.g. more than 500)

The screenshot shows the 'MID Server Cluster' configuration page for 'RomeMIDCluster'. At the top, there is a header bar with a back arrow and the cluster name. Below this, a blue informational banner states: 'MID Server Clusters enable multiple MID Servers with the appropriate capabilities to be grouped together. This is useful for load balancing and fail-over protection of MID Servers. [More Info](#)'. The configuration fields show 'Name' as 'RomeMIDCluster' and 'Type' as 'Load Balance'. Below these fields are 'Update' and 'Delete' buttons. A table below shows the list of MID servers included in the cluster. The table has columns for 'MID server', 'Status', 'Host OS', and 'Version'. The table title is 'MID server cluster = RomeMIDCluster'. The table contains two rows: 'MIDServer_Rome_01' with status 'Operational' and 'MIDServer_Rome_02' with status 'Operational'. Above the table is a filter bar with 'Includes MID Servers', an 'Order' dropdown, and a 'Search' input field.

MID server	Status	Host OS	Version
MIDServer_Rome_01	Operational		
MIDServer_Rome_02	Operational		

Communication Architecture – MID Server



MID Server Application Configuration - Detailed

Product	HEAP GB	Memory GB	CPU cores	HDD GB
Discovery	1	4	4	50
Service Mapping	1	4	4	50
Health Log Analytics	1	4	4	50
CPG	4	8	4	50
ACC	8	16	8	50
EM/OI	4	12	4	36

Installation

Using the ITOM Guided Setup



Guided Setup – MID Servers

The screenshot shows the 'ITOM Guided Setup' interface. On the left, a sidebar menu has 'ITOM Guided Setup' circled in blue. Below it, under 'ITOM License', 'License Summary' is also circled in blue. The main content area is titled 'IT Operations Management Guided Setup' and shows a progress bar at 0%. A card for 'MID Server' is circled in blue, with a 'Get Started' button below it also circled in blue. A text box explains that the MID Server runs as a Windows service or a UNIX daemon. To the right, a list of tasks is shown: 'Create MID User', 'Download & Install MID', and 'Validate MID'. Below this, another task list shows 'Select CI Data to Collect' and 'Quick-Start Discovery'. Three numbered callouts are present: 1. 'Click on a Task and complete' (pointing to the task list), 2. 'Assign the Task' (pointing to the 'Get Started' button), and 3. 'Mark Complete' (pointing to the task list).

ITOM Guided Setup

ITOM License

- License Summary
- License Daily Usage Count
- License Report
- Report ITOM Licensable CIs
- Licenses by CI Types

IT Operations Management Guided Setup

0%

MID Server

The MID Server runs as a Windows service or a UNIX daemon to facilitate communication and the movement of data between a ServiceNow instance and external applications, data sources, and services. Complete the activities in this category to create a user for the MID Server, download the installation package, and validate the MID Server after installation.

Status: Not Started

Get Started

Status: Locked

Preview

1. Click on a Task and complete

0 / 3 Tasks completed

- Create MID User
- Download & Install MID
- Validate MID

2. Assign the Task

Leading Practice:
Use guided setup to step through the initial configuration. Guided setup assists you with planning the roll-out of the product and performing the basic configuration to go live.

3. Mark Complete

0 / 2 Tasks completed

- Select CI Data to Collect
- Quick-Start Discovery

- Use the Guided Setup and to Create MID Servers
- Define necessary MID server users and roles and create accounts

Performance and Tuning



Performance Considerations

- The Speed of performance can be controlled by threads and memory
- These increase or decrease resources to the MID Server application
- Example:

While your local discovery may take a few hours to complete perhaps there is a need to increase that time as it is discovering through a bandwidth-sensitive network.

This is where threads and memory come into play.

Tuning Threads

- A thread is a thread of execution in a program. The MID Server JVM allows an application to have multiple threads of execution running concurrently.
- The MID Server default is for 25 threads to be executed at once.
- Increasing this number will increase the number of tasks (discovery probes) that it will do at any one time speeding up the discovery jobs.
- This can be found in the deployed application in the <agent_dir>\config.xml file
- Further Information on [ServiceNow docs](#)

Java Virtual Machine Memory Size

- The MID Server JVM memory default is 1 GB.
- This configuration can be found in the <MID Server name>\agent\conf\wrapper-override.conf configuration file.
- Increasing this number will allow the application to allocate itself more memory resources from the host. This is done in conjunction with increasing the threads as needed.

```
# OPTIONAL: Maximum Java Heap Size (in MB)  
wrapper.java.maxmemory=1024
```

- Further Information on [ServiceNow docs](#)

Performance Calculation

- There are many factors that determine the calculation, because of the complexity of this it requires the experience and assistance of ServiceNow Engineers to agree on the number of MID Servers required.
- A MID server can process about a probe per minute per thread, and there are about 7 probes per device on average.
- Both of those may vary, depending on the mix of devices (for example, it takes more probes to discover Windows servers than it does for routers) and the speed of your network communications and MID Server.
- If the IP ranges are unusually large, i.e., 100 or bigger and mostly empty, the MID server will spend a significant amount of time scanning non-responding IP addresses, slowing the performance.

Performance Calculation Example

- Suppose you have a location with about 2,500 devices (a mix of servers, network gear, printers, and workstations) all together. Further suppose that you wanted Discovery to do its job between 6 pm and midnight on Friday and 3 am on Saturday each week – a “Discovery window” of 6 hours.
- Below is a rough estimate to see if a single MID server configured to use 25 threads could achieve this goal.
 - Using the rules-of-thumb, discovering 2,500 devices would require about

$$7 * 2,500 = 17,500 \text{ probes}$$

With 25 threads, the MID server can process about 25 probes per minute.

Performance Calculation Example

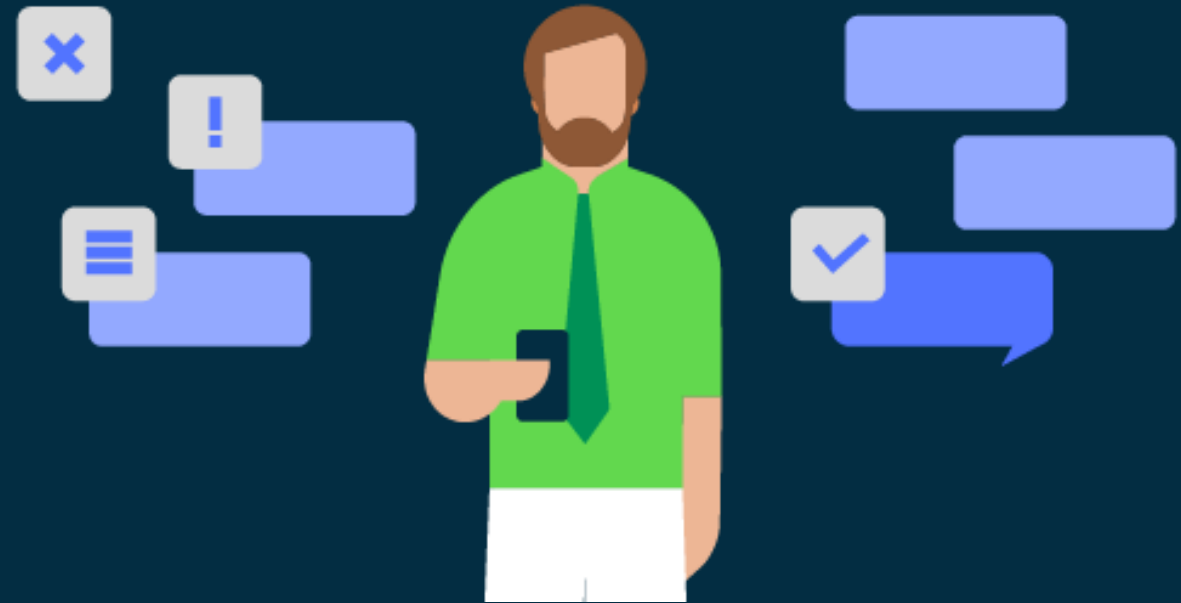
- Therefore, we can estimate that this work will take:

$17,500/25 = 700$ minutes, or about 11 hours

However, you need this work to be completed within the allotted window.

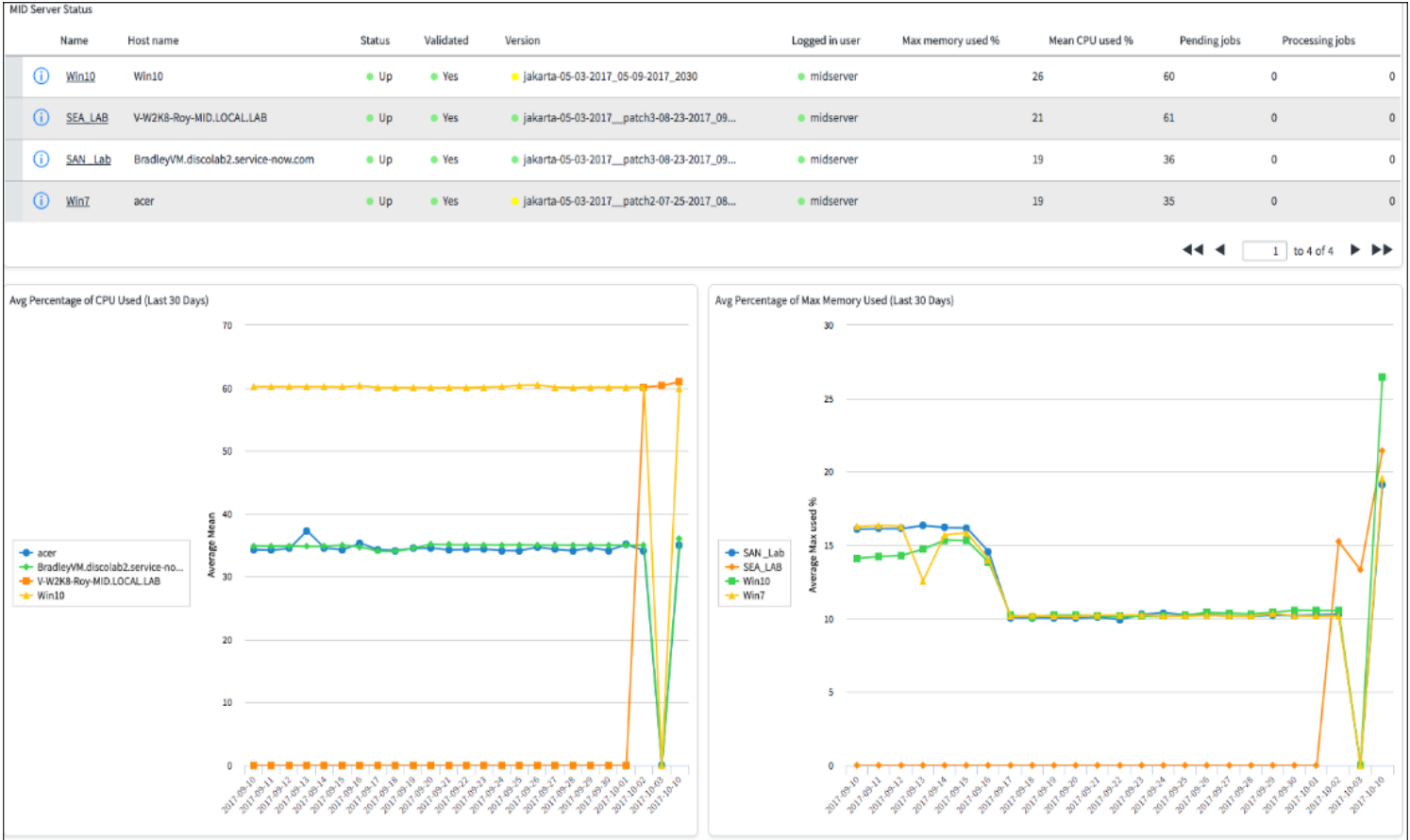
- Based on these numbers, you would want to increase the threads by 25 or you can tune those threads down where network utilization is a concern.
- There are other factors that determine the calculation like OS and type of active targets in addition to network latency.

Performance Monitoring



MID Server Performance Monitoring Dashboard

- Monitor the effects of increased threads and memory by:
1. Checking the MID Server dashboard at **MID Server > Dashboard**.
 2. A pair of charts on the default dashboard shows the average CPU and memory utilization for each MID Server.



MID Server Performance Monitoring Lists

Get more details on CPU and memory utilization in a pair of related lists on the MID Server form:

ECC Agent Scalar Metric->MID Server

ECC Agent Memory Metric->MID Server

The screenshot shows a 'Configuring MID Server form' dialog box. It has two main panes: 'Available' and 'Selected'. The 'Available' pane lists various metrics and configurations, including 'ECC Agent Scalar Metric->MID server' and 'ECC Agent Memory Metric->MID server'. The 'Selected' pane shows a hierarchy of fields, with the same two metrics selected at the bottom. Navigation arrows and 'Cancel'/'Save' buttons are present.

If you're using the default form, you'll need to use **Configure > Related Lists** to add these lists to the form

MID Server Performance Monitoring Lists (cont.)

The **scalar metric** shows the host's minimum, maximum, and average CPU utilization measured over a 10-minute period.

MID Server IssuesConfiguration Parameters (7)Supported Applications (1)IP Ranges (1)Capabilities (1)Included in ClustersExtension ContextsLogs (13)Threads (43)PropertiesScalar Metrics (312)Memory Metrics (154)

Scalar MetricsGo toNameSearch

1 to 20 of 312

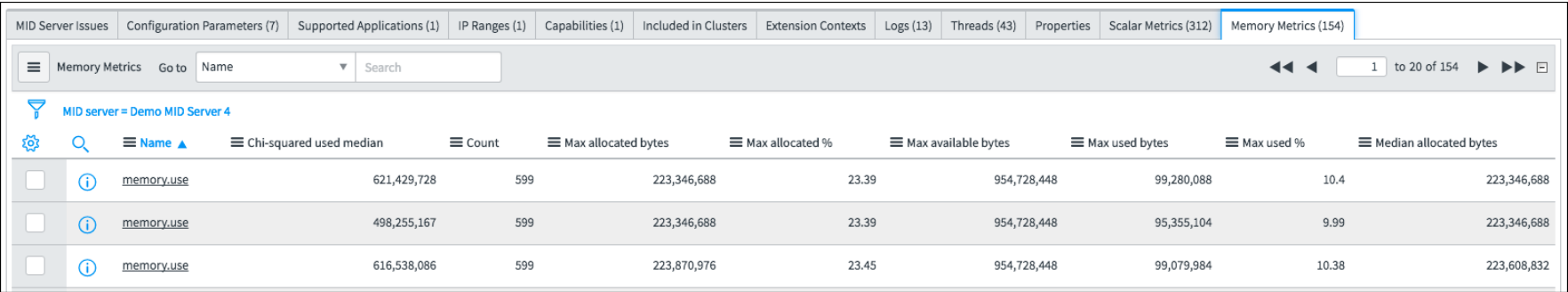
MID server = Demo MID Server 4

		Name ▲	Chi-squared mean	Chi-squared median	Count	Max	Mean	Median	Min	Domain
☐	i	cpu.use	676	875	599	74	36	33	33	global
☐	i	cpu.use	701	921	599	74	36	33	33	global
☐	i	cpu.use	647	841	599	75	36	33	33	global

MID Server Performance Monitoring Lists (cont.)

The **memory metric** shows the MID Server application's memory utilization measured over 10-minute periods. In particular:

- **Max available bytes** - the configured memory allocation for the MID Server as defined in the wrapper configuration file.
- **Max allocated bytes** - what the MID Server application has allocated itself from the configured available bytes.
- **Max used bytes** - memory resources actively being used by the MID Java Process.



The screenshot displays the 'Memory Metrics' tab in the ServiceNow MID Server Performance Monitoring interface. The interface includes a navigation bar at the top with various tabs like 'MID Server Issues', 'Configuration Parameters (7)', 'Supported Applications (1)', 'IP Ranges (1)', 'Capabilities (1)', 'Included in Clusters', 'Extension Contexts', 'Logs (13)', 'Threads (43)', 'Properties', 'Scalar Metrics (312)', and 'Memory Metrics (154)'. Below the navigation bar, there is a search bar and a table of memory metrics. The table has columns for 'Name', 'Chi-squared used median', 'Count', 'Max allocated bytes', 'Max allocated %', 'Max available bytes', 'Max used bytes', 'Max used %', and 'Median allocated bytes'. The table contains three rows of data, all for the 'memory.use' metric.

	Name	Chi-squared used median	Count	Max allocated bytes	Max allocated %	Max available bytes	Max used bytes	Max used %	Median allocated bytes
☐	memory.use	621,429,728	599	223,346,688	23.39	954,728,448	99,280,088	10.4	223,346,688
☐	memory.use	498,255,167	599	223,346,688	23.39	954,728,448	95,355,104	9.99	223,346,688
☐	memory.use	616,538,086	599	223,870,976	23.45	954,728,448	99,079,984	10.38	223,608,832

MID Server Performance Monitoring Lists (cont.)

As the MID Server:

- Sits idle there isn't going to be much memory consumption
- Is tasked to do work (like Discoveries) the Max used will start going up taking more allocated memory.

As more allocated memory gets utilized the more Java goes into its availability until it reaches the configured Max available.



Summary

- To get the most out of the MID Server application and host, start by increasing or decreasing the threads or the number of probes
- As that number increases so will the memory utilization. The more active the MID Server application, the host will need more of its own resources to manage the application
 - It is key that host performance is monitored as use of the MID server application increases. It is recommended to keep the host at about 80% total utilization at full scale for “standard” deployments.
- A base configured MID Server with 8GB of local memory should provide plenty of available resources to ‘give’ to the MID Server application as you increase threads.
- For cases like Discovery which are impacted by the number of targets to discover, increasing the amount of work a MID Server is tasked with will increase the network traffic as you are discovering more probes at a single time.

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